MA1220 – Calculus II – Solution

Q.1 Mark each statement True or False (**no justification required**) $(6 \times 1 = 6$ marks)

(a) True

(b) False

(c) False

(d) True

(e) False

(f) False

Q.2 Short Answer (Justification required): Answer the following. $(6 \times 1.5 = 9$ marks)

(a) Let $f(x) = 3x^2 - 12x + 12$. Find all $m > 1$ such that the average value of f on $[1, m]$ equals 3.

(b) If $g(2023) = \pi$, assume that g' is continuous, and $\displaystyle\int_{2023}^{2025} g'(x)\,dx = -\frac{\pi}{2}$, find $g(2025)$.

(c) If $f(x, y, z) = \dfrac{xy + yz^2 + xz^2}{x^2 + y^2 + z^4}$ for $(x, y, z) \neq (0, 0, 0)$, does $\lim\limits_{(x,y,z)\to(0,0,0)} f(x, y, z)$ exist?

(d) Determine whether the integral $\displaystyle\int_0^{\pi} \frac{\sin^2 x}{\sqrt{x}}\,dx$ converges or diverges.

(e) Find an equation of the tangent plane to the surface $z = e^{x-y}$ at the point $(2, 2, 1)$.

(f) Find the area enclosed by the curves $x = \sin y$ and $x = 1 - \cos y$ for $0 \leq y \leq \frac{\pi}{2}$.

Q.2 (Short Answer) — Solutions.

(a) *Given that the average value for* $f(x) = 3x^2 - 12x + 12$ *on the interval* $[1, m]$ *is* 3. Therefore,

$$\frac{1}{m-1}\int_1^m (3x^2 - 12x + 12)\,dx = 3 \qquad \text{1 mark}$$

On integrating, we get

$$\frac{m^3 - 6m^2 + 12m - 7}{m-1} = 3 \implies (m-1)^2(m-4) = 0.$$

With $m > 1$, this implies $\boxed{m = 4}$. ($\frac{1}{2}$ mark)

(b) Since g' is continuous on $[2023, 2025]$, by the Fundamental Theorem of Calculus,

$$g(2025) - g(2023) = \int_{2023}^{2025} g'(x)\,dx = -\frac{\pi}{2}.$$

Given $g(2023) = \pi$, it follows that

$$g(2025) = \pi - \frac{\pi}{2} = \boxed{\frac{\pi}{2}}.$$

(1.5 marks)

(c) *Limit of* $f(x,y,z) = \dfrac{xy + yz^2 + xz^2}{x^2 + y^2 + z^4}$ *as* $(x,y,z) \to (0,0,0)$. Along $y = x$, $z = 0$: $f = \dfrac{x^2}{x^2 + x^2} = \frac{1}{2} \to \frac{1}{2}$. Along $y = 0$, $z = 0$: $f = 0$. Since different paths give different limits, the limit $\boxed{\text{does not exist}}$. (1.5 marks)

(d) *The improper integral* $\displaystyle\int_0^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx$ *is defined as follows:*

$$\int_0^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx = \lim_{t\to 0^+} \int_t^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx \qquad \tfrac{1}{2} \text{ mark}$$

Then

$$\lim_{t\to 0^+} \int_t^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx \;\le\; \lim_{t\to 0^+} \int_t^\pi \frac{1}{\sqrt{x}}\,dx \qquad \tfrac{1}{2} \text{ mark}$$

$$\lim_{t\to 0^+} \int_t^\pi \frac{1}{\sqrt{x}}\,dx = \lim_{t\to 0^+} \Big[2\sqrt{x}\Big]_t^\pi = \lim_{t\to 0^+} \big(2\sqrt{\pi} - 2\sqrt{t}\big) = 2\sqrt{\pi} < \infty. \qquad \tfrac{1}{2} \text{ mark}$$

Hence the improper integral $\boxed{\text{converges}}$.

Common mistakes. In general, many students made mistakes in this question and lost partial or full marks. Some did not write the improper integral as $\int_0^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx = \lim_{t\to 0^+} \int_t^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx$ (this step is absolutely necessary). Others wrote the definition and compared $\sin^2 x$ with a suitable function but did not evaluate the bounding integral. Sometimes people wrote, "by the comparison theorem, the integral is convergent," without properly stating the exact comparison. There were also other variants of mistakes (e.g., incorrect use of a Riemann sum).

(e) *Tangent plane to* $z = e^{x-y}$ *at* $(2,2,1)$. Here $f_x = e^{x-y}$, $f_y = -e^{x-y}$; at $(2,2)$: $f_x = 1$, $f_y = -1$. ($\frac{1}{2}$ mark)

Thus, the equation of the tangent plane is

$$z - 1 = 1 \cdot (x-2) + (-1)\cdot(y-2) = x - y \quad \Rightarrow \quad \boxed{z = 1 + x - y}.$$

(1 marks)

(f) *Area enclosed by* $x = \sin y$ *and* $x = 1 - \cos y$, $0 \le y \le \frac{\pi}{2}$. On $[0, \frac{\pi}{2}]$, $\sin y \ge 1 - \cos y$, so

$$A = \int_0^{\pi/2} \big(\sin y - (1 - \cos y)\big)\,dy \qquad 1 \text{ mark}$$

Thus, we get

$$A = \Big[-\cos y + \sin y - y \Big]_0^{\pi/2} = \boxed{2 - \frac{\pi}{2}}. \qquad \tfrac{1}{2} \text{ mark}$$

Q.3 Answer the following. (14 marks)

(a) Find the area of the region bounded by the graphs of $y = \sin x$ and $y = x\sin(x^2)$ for $0 \le x \le 1$. (3 marks)

(b) Find the volume V obtained by rotating the region enclosed by the curves $y = 2\sqrt{x}$ and $y = x$ about the line $x = -2$. (*Extra note: Do not evaluate the integral; writing the correct integral expression for V is sufficient.*) (3 marks)

(c) Find the area of the surface obtained by rotating the curve $x = \sqrt{a^2 - y^2}$ for $0 \le y \le \frac{a}{2}$ about the x-axis. (4 marks)

(d) Let $f : [a, b] \to \mathbb{R}$ be continuous and define

$$g(x) = \int_a^x f(t)\, dt \quad (a \le x \le b).$$

Prove that g is differentiable on (a, b). (4 marks)

Solutions.

(a) *Area between $y = \sin x$ and $y = x\sin(x^2)$ on $0 \le x \le 1$.* Since $0 \le x^2 \le x \le 1$ and sin is increasing on $[0, 1]$, we have $\sin(x^2) \le \sin x$; also $0 \le x \le 1$ gives $x\sin(x^2) \le \sin x$. Hence the area is

$$A = \int_0^1 \big(\sin x - x\sin(x^2)\big)\, dx = \int_0^1 \sin x\, dx - \int_0^1 x\sin(x^2)\, dx. \qquad 2 \text{ marks}$$

Now

$$\int_0^1 \sin x\, dx = 1 - \cos 1, \qquad \int_0^1 x\sin(x^2)\, dx = \tfrac{1}{2}\int_0^1 \sin u\, du = \tfrac{1}{2}(1 - \cos 1).$$

Therefore

$$A = \Big(1 - \cos 1\Big) - \tfrac{1}{2}\Big(1 - \cos 1\Big) = \boxed{\tfrac{1}{2}\big(1 - \cos 1\big)}. \qquad 1 \text{ mark}$$

Common mistakes. Note that the area is not defined as

$$A = \left| \int_0^1 \big(\sin x - x\sin(x^2)\big)\, dx \right|.$$

Other mistakes include writing

$$A = \int_0^1 \big(x\sin(x^2) - \sin x\big)\, dx.$$

Some students evaluated the integral incorrectly.

(b) *Volume by rotating the region enclosed by $y = 2\sqrt{x}$ and $y = x$ about $x = -2$.*

Setting $2\sqrt{x} = x$ and squaring both sides gives $4x = x^2$ or $x(x-4) = 0$, so the two curves intersect at $x = 0$ and $x = 4$. (1 mark)

Volume by the washer method. Rotating the region enclosed by $y = 2\sqrt{x}$ and $y = x$ about $x = -2$ produces a solid whose cross-sections are washers with outer radius $R = y - (-2) = y + 2$ and inner radius $r = \frac{y^2}{4} - (-2) = \frac{y^2}{4} + 2$. The volume of the solid of revolution is

$$V = \pi \int_0^4 \left[(y+2)^2 - \left(\tfrac{y^2}{4} + 2\right)^2 \right] dy \qquad \text{2 marks}$$

Volume by the cylindrical shells method. Using cylindrical shells about the vertical line $x = -2$:

$$V = 2\pi \int_0^4 (\text{radius})(\text{height})\, dx = 2\pi \int_0^4 (x+2)\left(2\sqrt{x} - x\right) dx. \qquad \text{2 marks}$$

Note. One can find the volume in the previous question by either of the above two methods. Both answers are acceptable.

Common mistakes. In this question, the volume V is obtained by rotating the region enclosed by the curves about the line $x = -2$. Some students used $x = 0$ or $y = 0$ instead. You lose 2 marks for that.

(c) *Surface area when rotating $x = \sqrt{a^2 - y^2}$, $0 \le y \le \frac{a}{2}$, about the x-axis (assume $a > 0$).* For rotation about the x-axis using y as the parameter,

$$S = 2\pi \int_0^{a/2} y \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy, \quad \text{where} \quad x = \sqrt{a^2 - y^2}. \qquad \text{2 marks}$$

Here

$$\frac{dx}{dy} = -\frac{y}{\sqrt{a^2 - y^2}} \quad \Rightarrow \quad \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \frac{a}{\sqrt{a^2 - y^2}}.$$

Thus

$$S = 2\pi a \int_0^{a/2} \frac{y}{\sqrt{a^2 - y^2}}\, dy. \qquad \text{1 mark}$$

Set $u = a^2 - y^2$, then $du = -2y\, dy$, and we get

$$S = 2\pi a \left[-\sqrt{a^2 - y^2} \right]_0^{a/2} = 2\pi a \left(a - \frac{\sqrt{3}}{2} a \right) = \boxed{\pi a^2 \left(2 - \sqrt{3}\right)}. \qquad \text{1 mark}$$

Common mistakes. In the question, rotation is about the x-axis. If you choose the wrong axis or line, you will lose marks. Some students also wrote an incorrect formula for the surface area.

(d) We shall show g is differentiable at an arbitrary point $x \in (a, b)$. Assume $h > 0$ and $x, x+h \in (a, b)$. Then

$$g(x+h) - g(x) = \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt = \int_x^{x+h} f(t)\,dt.$$

Thus,

$$\frac{g(x+h) - g(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt. \qquad \text{1 mark}$$

Since f is continuous on $[x, x+h]$, there exist $u, v \in [x, x+h]$ with $f(u) = m$ (minimum) and $f(v) = M$ (maximum). Using a standard property of definite integrals,

$$mh \le \int_x^{x+h} f(t)\,dt \le Mh,$$

i.e.,

$$f(u) \le \frac{1}{h}\int_x^{x+h} f(t)\,dt \le f(v). \qquad \text{1 mark}$$

Hence

$$f(u) \le \frac{g(x+h) - g(x)}{h} \le f(v).$$

Let $h \to 0$. Then $u \to x$ and $v \to x$; by continuity, $f(u), f(v) \to f(x)$. Taking limits,

$$f(x) \le \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} \le f(x). \qquad \text{1 mark}$$

By the squeeze theorem,

$$\lim_{h\to 0} \frac{g(x+h) - g(x)}{h} = f(x) \quad \text{(limit exists)}. \qquad \text{1 mark}$$

Thus, g is differentiable on (a, b).

Q.4 Answer the following. (16 marks)

(a) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(0,0) = 0$ and

$$f(x,y) = \begin{cases} x\sin\dfrac{1}{x} + y\sin\dfrac{1}{y}, & \text{if } x \neq 0 \text{ and } y \neq 0, \\ x\sin\dfrac{1}{x}, & \text{if } x \neq 0 \text{ and } y = 0, \\ y\sin\dfrac{1}{y}, & \text{if } y \neq 0 \text{ and } x = 0. \end{cases}$$

Show that f is continuous at $(0,0)$ (give an ε–δ proof). Show that the partial derivatives of f do not exist at $(0,0)$.

(b) Let $f(x,y) = x^3 + xy^3$. Find all unit directions at the point $(1,1)$ for which the directional derivative is 0.

(c) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x,y) = x^3$. Find the local maximum and minimum values and the saddle point(s) of f (if they exist).

(d) Find the extreme values of $f(x,y) = 2x+4y$ subject to the constraint $g(x,y) = x^2+y^2-5 = 0$.

Solutions.

(a) **Checking continuity of f at the origin $(0,0)$.**

Notice that for all $x \neq 0$, $y \neq 0$,

$$|x \sin \tfrac{1}{x}| \leq |x|, \qquad |y \sin \tfrac{1}{y}| \leq |y|.$$

Hence for all (x,y),

$$|f(x,y)| \leq |x| + |y| \leq \sqrt{2}\,\sqrt{x^2+y^2}. \qquad \text{1 mark}$$

Let $\varepsilon > 0$ and choose $\delta = \varepsilon/\sqrt{2}$. If $0 < \sqrt{x^2+y^2} < \delta$, then

$$|f(x,y) - 0| \leq \sqrt{2}\,\sqrt{x^2+y^2} < \sqrt{2}\,\delta = \varepsilon. \qquad \text{1 mark}$$

Therefore, $\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$, so f is continuous at $(0,0)$.

Checking existence of partial derivatives of f at the origin $(0,0)$.

By definition, the partial derivative with respect to x at $(0,0)$ is

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{h \sin(1/h)}{h} = \lim_{h\to 0} \sin\left(\frac{1}{h}\right),$$

which does *not* exist because $\sin(1/h)$ oscillates between -1 and 1 infinitely often as $h \to 0$. Thus, the partial derivative of f at the origin $(0,0)$ with respect to x does not exist. (1 mark)

Similarly, the partial derivative with respect to y at $(0,0)$ is

$$f_y(0,0) = \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k\to 0} \sin\left(\frac{1}{k}\right),$$

which does *not* exist. Thus, the partial derivative of f at the origin $(0,0)$ with respect to y does not exist. (1 mark)

Common mistakes.

- Some students wrote that for all $x \neq 0$, $y \neq 0$, the function $x \sin \dfrac{1}{x} + y \sin \dfrac{1}{y}$ is a sum of continuous functions and then "approached $(0,0)$," hence $f(x,y)$ is continuous. This is

incomplete and cannot be accepted as a correct answer. One needs to show that for all $(0,0) \neq (x,y) \in \mathbb{R}^2$ (including along coordinate axes), $f(x,y) \to 0$ as $(x,y) \to (0,0)$.

- Many tried to bound the function but missed absolute-value signs.
- The function is defined by three cases; hence one cannot directly compute partial derivatives of the piecewise formula first and then take the limit to $(0,0)$. This approach is incorrect.

(b) Let $f(x,y) = x^3 + xy^3$. Then

$$\nabla f(x,y) = \big(3x^2 + y^3,\ 3xy^2\big) \quad \Rightarrow \quad \nabla f(1,1) = (4,3). \qquad \text{1 mark}$$

Recall two theorems from the lecture:

- If f is differentiable in x and y, then for any unit vector $\mathbf{u} = \langle a, b\rangle$,
$$D_{\mathbf{u}} f(x,y) = f_x(x,y)\, a + f_y(x,y)\, b.$$
- If $f_x(x,y)$ and $f_y(x,y)$ exist and are continuous on an open disk $\mathcal{D}$, then f is differentiable on $\mathcal{D}$.

Using these, we conclude that f is differentiable, and hence for a unit vector $\mathbf{u} = (u_1, u_2)$, the directional derivative at $(1,1)$ is

$$D_{\mathbf{u}} f(1,1) = \nabla f(1,1) \cdot \mathbf{u} = 4u_1 + 3u_2. \qquad \text{1 mark}$$

Thus, $D_{\mathbf{u}} f(1,1) = 0$ iff $4u_1 + 3u_2 = 0$ with $u_1^2 + u_2^2 = 1$. (1 mark)

From here, conclude that

$$\boxed{\mathbf{u} = \big(-\tfrac{3}{5},\ \tfrac{4}{5}\big) \text{ or } \big(\tfrac{3}{5},\ -\tfrac{4}{5}\big)}. \qquad \text{1 mark}$$

(c) Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) = x^3$.

Compute first derivatives:

$$f_x = 3x^2, \qquad f_y = 0.$$

Then $f_x = f_y = 0$ implies $x = 0$ (with any y). Hence *every* point on the y-axis is a critical point, i.e., the set of critical points is $\{(0,y) : y \in \mathbb{R}\}$. (1 mark)

Second derivative test.

$$f_{xx} = 6x, \quad f_{xy} = 0, \quad f_{yy} = 0 \qquad \text{implies} \qquad D = f_{xx} f_{yy} - f_{xy}^2 = 0. \qquad \text{1 mark}$$

So the second derivative test is inconclusive at *every* critical point.

To check behavior at an arbitrary point $(0, y_0)$:
First, approach $(0, y_0)$ via (h, y_0) for some $h > 0$. Then $f(h, y_0) = h^3 > 0$. (1 mark)

Now approach $(0, y_0)$ via (k, y_0) for some $k < 0$. Then $f(k, y_0) = k^3 < 0$. (1 mark)

Hence, there are *no* local maxima or minima; every point on the y-axis is a saddle point:

$$\boxed{\text{No local maxima/minima; saddle points are all } (0, y),\ y \in \mathbb{R}.}$$

Common mistakes.

- Some treated $f(x, y) = x^3$ as a single-variable case. This is incorrect.
- If you did not mention the critical points even after finding f_x and f_y, you get 0.5 mark.
- Many wrote only $(0, 0)$ as a critical point. You lose partial marks here (you still get 0.5 mark).
- The critical points are the entire y-axis. The analysis for maxima or minima should be done for arbitrary critical points on the y-axis, not just at $(0, 0)$.
- A graphical sketch alone may not give full justification at critical points; you need to write some argument to support your conclusion.

(d) Maximise/minimise $f(x, y) = 2x + 4y$ subject to $g(x, y) = x^2 + y^2 - 5 = 0$.

Using the method of Lagrange multipliers, set $\nabla f = \lambda \nabla g$. Then

$$\nabla f = (2, 4) = \lambda(2x, 2y) \qquad \text{implies} \qquad (1, 2) = \lambda(x, y).$$

Thus, (x, y) is proportional to $(1, 2)$; write $(x, y) = (k, 2k)$ for some k. (2 marks)

The constraint gives $k^2 + (2k)^2 = 5 \Rightarrow 5k^2 = 5 \Rightarrow k = \pm 1$, so the candidates are $(1, 2)$ and $(-1, -2)$. (1 mark)

Evaluating at these points, we get $f(1, 2) = 10$ and $f(-1, -2) = -10$. Thus, we conclude that

$$\boxed{f_{\max} = 10 \text{ at } (1, 2); \quad f_{\min} = -10 \text{ at } (-1, -2).} \qquad 1 \text{ mark}$$

Common mistakes. Many students adopted different approaches, which is fine. Here are some common mistakes.

- For instance, using the Cauchy–Schwarz inequality, one may write

$$|f(x, y)| = |2x + 4y| \le \sqrt{2^2 + 4^2}\,\sqrt{x^2 + y^2} = \sqrt{20}\,\sqrt{5} = \sqrt{100} = 10.$$

Thus, $-10 \le f(x, y) \le 10$. Many stopped here, claiming the maximum is 10 and the minimum is -10 without justifying whether equality holds. Similar incomplete solutions were written using polar coordinates as well: $x = 5\cos\theta$ and $y = 5\sin\theta$. You will get partial marks for this step (2 marks). One has to show equality holds and at which points.

- To complete the solution, one has to show equality holds when (x, y) and $(2, 4)$ are proportional; that is, $(x, y) = k(2, 4)$ for some constant k. Then $x = 2k$, $y = 4k$. Substituting these into the constraint $x^2 + y^2 = 5$ gives $k = \pm\frac{1}{2}$. From here, you get the two points $(-1, -2)$ and $(1, 2)$. Then $f(1, 2) = 10$ is the maximum and $f(-1, -2) = -10$ is the minimum.